

A second electron with opposite spin is attracted by the deformation. The two electrons form a pair that, with others, can create a Bose-Einstein condensate (see pp.78–79).

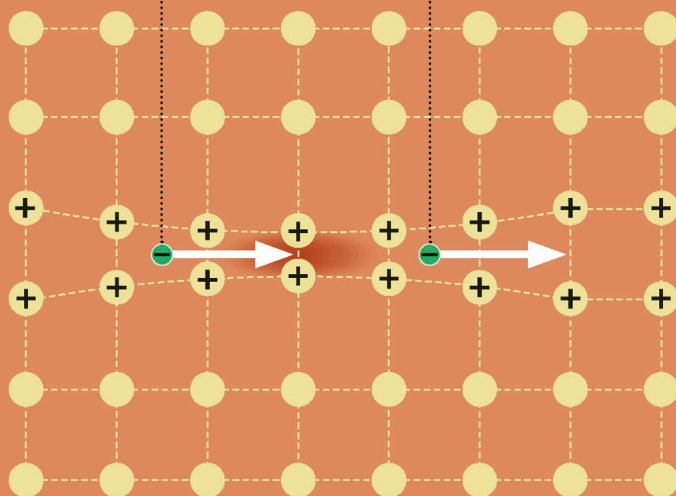
ELECTRON TWO

As an electron moves through a superconductor at low temperatures it attracts positive ions toward it.

ELECTRON ONE

BCS model

The Bardeen-Cooper-Schrieffer model is a long-standing and powerful way of explaining most types of superconductivity.



POSITIVELY CHARGED ION LATTICE

ENDLESS CHARGE

In normal electrical conductors, energy is lost to resistance—interactions between the electron particles that transmit current and their surroundings. However, at very low temperatures quantum behavior leads to an effect called superconductivity, in which current flows without resistance through certain materials. Electrons in a superconductor travel in Cooper pairs that exhibit the properties of bosons (see pp.78–79) rather than fermion particles, allowing them to flow without friction in a similar way to superfluids.